



Özge Drama Berkenkamp

Senior Robotics Control Software Engineer
Humanoids R&D



Munich, Germany



<https://oezgedrama.me>



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Motivation

I am passionate about control, optimization, reinforcement learning, and simulation to solve complex challenges in humanoids and promote modern software practices across the team.

Languages

- Turkish: Native
- English: Fluent
TOEFL IBT: 113
- German: Level B2-C1
Goethe Cert. B2
- Japanese: Level A2-B1

Programming

- C++, Python
- PyTorch, TensorFlow, IsaacLab
- ROS2, LN
- CI/CD, docker, git, (c)make
- MATLAB, Simulink, LabVIEW
- Simulation: MuJoCo, PyBullet
- SolidWorks, SolidCAM, AutoCAD
- Comsol Multiphysics
- Xilinx ISE, dSPACE
- SIMATIC Manager (PLC)

Work Experience

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|-----------|---|-----------------------|
| 2024-... | Senior Robotics Control Software Engineer
Agile Robots SE | Munich, Germany |
| | • Humanoid control: RL-based locomotion and loco-manipulation policy development for the Agile ONE humanoid | |
| 2024 | Senior Robotics Software Engineer
Angsa Robotics | Munich, Germany |
| | • Algorithms for wheel slip&stuck detection, and overcurrent protection | |
| 2021-2023 | Robotics Control Engineer
Franka Emika GmbH | Munich, Germany |
| | • Motor & Joint Control: Optimal gain tuning for PMSM FOC drives and joint full-state feedback controllers | |
| | • 7-DOF Robot Control: Evaluation of gravity compensation accuracy under model uncertainty and crosstalk torques | |
| | • Modeling & Identification: Python framework for symbolic robot modeling and physically feasible dynamic parameter identification | |
| | • Robot Dimensioning: Joint torque estimation for custom robot design | |
| 2015-2017 | Research and Development Engineer
Maxon Motor AG | Sachseln, Switzerland |
| | • Modeling, simulation, and improvement of DC brush motors | |
| | • Research on DC motor loss reduction | |
| | • Software and testbed development for measuring motor properties | |

Education

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|-----------|---|---------------------|
| 2017-2021 | Ph.D. in Mechanical Engineering
University of Stuttgart
Max Planck Institute for Intelligent Systems, Dynamic Locomotion Group
Thesis: Control Mechanisms for Postural Stability and Trunk Motion in Bipedal Running: A Study for Humans, Avians, and Bipedal Robots | Stuttgart, Germany |
| 2012-2015 | M.Sc. in Robotics, Systems and Control
ETH Zurich
Thesis: Trajectory Optimization for Fall Recovery of a Quadruped Robot | Zurich, Switzerland |
| 2008-2012 | B.Sc. in Mechatronics Engineering
Sabancı University with 100% Merit Scholarship
Ranking: 4 th /51
Thesis: ZMP-based Trajectory Generation for a Quadruped Robot | İstanbul Turkey |

Practical Experience

Internships

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| 2013
Sep– Feb | Robert Bosch GmbH
• Real-time PMSM Field-Oriented Control (FOC) implementation
• Electromagnetic interference reduction using spread spectrum technique | Schwieberdingen, Germany |
| 2011
Jun–Sep | DMS (Digitale Mess- und Steuersysteme) AG
• HVAC system design for industrial automation | Stuttgart, Germany |
| 2010
Aug–Sep | Figes A.Ş.
• Multidomain modeling and control acceleration via Simulink/Simscape | Bursa, Turkey |
| 2010
Jun–Aug | Bosch San. ve Tic. A.Ş.
• Obliqueness measurement tool design for HDEV fuel injector valve body | Bursa, Turkey |

Hobbies

- Dancing: ballet, Lindy Hop
- Music: piano, singing
- Hiking

Summer schools

2020	MEMMO: Memory of motion summer school	Virtual
2017	PhD summer school on learning systems	ETH Zurich, Switzerland
2015	Robotics summer school	Tohoku University, Sendai, Japan

Seminars

2018-2021	IMPRS-IS soft skill seminars • Intercultural communication • Poster design and presentation • Voice & language: How to communicate more easily and effectively • Media training: Science communication • Leadership training	Stuttgart, Germany
2016	Embedded control and monitoring using LabVIEW National Instruments	Switzerland
2015	Foundations and applications of small electrical motors WDI Wissensforum, Technical University of Ilmenau	Germany
2011-2012	Proficiency in mechatronics education FESTO and Boğaziçi University	İstanbul, Turkey
2009	IMES AutoCAD education	Bursa, Turkey

Publications

Journal Papers

2020	Virtual point control for step-down perturbations and downhill slopes in bipedal running Drama, Ö. and Badri-Spröwitz, A. Frontiers in Bioengineering and Biotechnology
2020	Postural stability in human running with step-down perturbations: an experimental and numerical study Drama, Ö., Vilemeyer J., Badri-Spröwitz, A., and Müller R. Royal Society Open Science
2020	Trunk pitch oscillations for energy trade-offs in bipedal running birds and robots Drama, Ö. and Badri-Spröwitz, A. Bioinspiration & Biomimetics

Conference Papers

2019	Trunk pitch oscillations for joint load redistribution in humans and humanoid robots Drama, Ö. and Badri-Spröwitz, A. IEEE-RAS International Conference on Humanoid Robots
2011	ZMP reference trajectory generation with preview control for a quadruped robot Fidan K., Akbaş T., Eskimez E., Özel S., Adak K., Drama Ö., Konukoğlu M., Yılmaz G., and Erbatur K. Turkey Automation Conference

Presentations and Abstracts

2017	Linking mechanics and learning Heim S., Grimminger F., Drama Ö., and Badri-Spröwitz A. Dynamic Walking Conference
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